



A High-Resolution adaptive beamforming algorithm Real-Time implemented by FPGA for ultrasound imaging

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ABSTRACT

The delay and sum (DAS) beamforming algorithm commonly employed in ultrasound imaging systems suffers from insufficient resolution and contrast, prompting many adaptive beamforming algorithms to be proposed for improving image quality. However, some high performance beamforming algorithms are difficult to implement in real-time imaging systems due to the heavy calculating resources required. Therefore, this paper firstly proposes a low complexity signal range-standard deviation factor (SRSF), which can effectively suppress the off-axis signals, improve resolution, and avoid damaging the speckle background. Secondly, combining SRSF and multiphase apodization with cross-correlation (MPAX), the SRSF-MPAX beamformer is obtained to enhance the cyst detection capability while maintaining the resolution advantage. Finally, the implementation scheme of SRSF-MPAX on FPGA is introduced in detail. According to the results of the simulation, *human heart* experiment and *ultrasound phantom* experiment, compared with DAS, the full-width at half-maximum (FWHM) of SRSF-MPAX is decreased by 71.10%, 49.49%, and 63.93% respectively, and the contrast ratio (CR) of SRSF-MPAX is improved by 51.28%, 43.93%, and 53.19% respectively. The proposed SRSF-MPAX beamformer consumes 55%, 68%, and 96% of the logic, memory, and DSP resources on FPGA (EP4SE360F35I3), respectively. The overall imaging throughput of the SRSF-MPAX beamformer on FPGA is 2289.4MByte/s input and 95.4MByte/s output, with about 4.03W theoretical power dissipation.

1. Introduction

As one of the four major medical imaging technologies, ultrasound imaging has the advantages of safe, non-invasive, and rapid detection [1]. The detection objects of ultrasound imaging are various, such as the human heart, liver, thyroid, and blood vessels [2,3]. The imaging modes mainly include focused imaging, synthetic aperture imaging, and plane wave imaging [4,5]. The delay and sum (DAS) beamformer is the simplest and most widely used ultrasound imaging method, which involves time-aligning and summing the received echoes [6]. However, due to the wide mainlobe and high sidelobes, the DAS beamformer has low resolution and cannot suppress clutter within anechoic cysts.

To improve the resolution of ultrasound images, Capon proposed the minimum variance (MV) beamformer [7], and many improvements have been introduced to make it suitable for broadband ultrasound [8,9]. It should be noted that the MV beamformer requires an inversion of the covariance matrix, which leads to high computational complexity

[10]. In addition, to suppress clutter within anechoic cysts, beamformers that use the amplitude and phase coherence of echo signals have been proposed, such as coherence factor (CF) [11], sign coherence factor (SCF) [12], phase coherence factor (PCF) [13] and generalized coherence factor (GCF) [14]. However, owing to the over-suppression of weakly coherent signals, these beamformers are prone to introducing dark-region artifacts and deteriorating speckle background quality. In contrast to CF-based beamformer, the dual apodization with cross-correlation (DAX) beamformer suppresses clutter by normalized cross-correlation coefficient of different grating lobes [15,16]. Furthermore, the multiphase apodization with cross-correlation (MPAX) beamformer is proposed to enhance the robustness of DAX [17].

In addition to theoretical research, implementing adaptive beamformers in real-time imaging systems is also a current research hotspot [18,19]. Adaptive beamformer implementations are mainly based on central processing unit (CPU), graphics processing unit (GPU) [20], and field programmable gate array (FPGA) [21]. In real-time ultrasound

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imaging, a large number of echo signals must be processed in a short period. The CPU cannot rapidly process array echo signals through its serial processing architecture [22]. Although the GPU has a powerful parallel processing capability, it relies on the support of the PC, resulting in high cost and bloated volume [23]. It is worth noting that the FPGA can provide a flexible way to design digital circuits through its reconfigurable and programmable features, which are well suited to parallel processing of array signals [24]. Therefore, many researchers have focused their attention on the implementation of FPGA-based adaptive beamformers [25,26]. Compared with beamformers that involve inverse operations, CF and SCF beamformers are easier to implement on FPGA. However, the CF algorithm for real-time imaging still suffers from dark-region artifacts and insufficient resolution. Therefore, it is valuable to propose a beamforming algorithm with good performance and low computational complexity that can be implemented on FPGA for real-time ultrasound imaging.

In this paper, a signal range-standard deviation factor (SRSF) is proposed to improve lateral resolution without losing the speckle background quality. Then, the SRSF-MPAX beamforming algorithm is obtained by combining the MPAX and SRSF algorithms, which can simultaneously improve the resolution and clutter suppression capabilities. The simulation and *human heart* data acquired by the Verasonics P4-2v also are used to validate the performance of SRSF-MPAX. In addition, the SRSF-MPAX beamformer is achieved on the FPGA-based ultrasound imaging platform developed by our research group for real-time imaging. According to the results of Quartus software, it can be found that compared with DAS, the proposed SRSF-MPAX beamformer increases the resource consumption on the FPGA, but the overall image quality is improved.

The content of this paper is organized as follows: section II contains the conventional beamformers, and section III illustrates the proposed SRSF-MPAX. The section IV is the results of simulation and experiments. The section V and section VI are discussion and conclusion respectively.

2. Background

2.1. Delay and sum beamformer

For an array with M elements, the output of DAS beamformer is as follows:

$$Y_{DAS}(n) = \rho^H(n)x(n) \quad (1)$$

where n is the time index, and $x(n)$ is the delayed signal. $\rho^H(n)$ is the weight vector. $(\cdot)^H$ represents a conjugate transpose operation.

2.2. Coherence factor and sign coherence factor beamformers

The weight vectors of coherence factor (ρ_{CF}) and sign coherence factor (ρ_{SCF}) can be expressed as follows:

$$\begin{cases} \rho_{CF}(n) = \left| \sum_{m=1}^M x_m(n) \right|^2 / \left(M \sum_{m=1}^M |x_m(n)|^2 \right) \\ \rho_{SCF}(n) = 1 - \sigma = 1 - \sqrt{1 - \left(\frac{1}{M} \sum_{m=1}^M \text{sign}(x_m(n)) \right)^2} \end{cases} \quad (2)$$

where x_m is the delayed echo signal of the m th element, σ is the standard deviation of the signs of the delayed echo signal, and $\text{sign}(\bullet)$ is the sign operation. When $x_m(n) \geq 0$, $\text{sign}(x_m(n))$ is 1; when $x_m(n) < 0$, $\text{sign}(x_m(n))$ is -1 .

The output of CF and SCF beamformers are as follows:

$$\begin{cases} Y_{CF}(n) = \rho_{CF}(n) \times Y_{DAS}(n) \\ Y_{SCF}(n) = \rho_{SCF}(n) \times Y_{DAS}(n) \end{cases} \quad (3)$$

2.3. Multiphase apodization with cross-correlation beamformer

The MPAX beamformer generates signal pairs with different grating lobes through multiple complementary sinusoidal phase apodization functions. Then a weighting matrix using the normalized cross-correlation between the signal pairs is constructed to effectively suppress reverberation clutter. Suppose N sets of complementary sinusoidal phase apodizations with the spatial frequency of $\{f_0^1, f_0^2, \dots, f_0^N\}$ are used to obtain the signal pairs y_{1i} and y_{2i} ($i = 1, 2, \dots, N$). At the transmitting focal depth, the composite pulse-echo field $\psi_{\omega, PA}$ obtained from DAS beamforming with sinusoidal phase apodization can be represented by Eq.(4), which can predict the distance between the generated grating lobes and mainlobe [17,27].

$$\begin{aligned} \psi_{\omega, PA}(x, z) &= (e^{jkz} / (j\lambda z))^2 (2a)^2 \text{sinc}[2\pi a(x/(\lambda z))] \\ &\times \sum_{q=-\infty}^{\infty} J_q(m_p/2) \text{sinc}[(2\pi a/(\lambda z))(x - qf_0\lambda z)] \end{aligned} \quad (4)$$

where q is the order, λ is the wavelength, and J_q is the first kind of Bessel function. $k = 2\pi/\lambda$ is the wavenumber, and m_p (in radians) is the peak-to-peak phase delay. $2a$ is the length of a one-dimensional array, ω is the frequency and (x, z) is the field point.

According to Eq.(4), different f_0 will generate grating lobe pairs with distinct information, which can better suppress clutter by normalized cross-correlation. Then, the weighted matrix of MPAX (ρ_{MPAX}) is calculated as follows [28]:

$$\rho_{MPAX}(n, s) = \max \left(\frac{1}{N} \sum_{i=1}^N \frac{\sum_{k=s-K}^{s+K} \sum_{j=n-A}^{n+A} y_{1i}(k, j) y_{2i}(k, j)}{\sqrt{\sum_{k=s-K}^{s+K} \sum_{j=n-A}^{n+A} y_{1i}(k, j)^2} \sqrt{\sum_{k=s-K}^{s+K} \sum_{j=n-A}^{n+A} y_{2i}(k, j)^2}}, \varepsilon \right) \quad (5)$$

where ε is the threshold value, $2A + 1$ is the length of samples, and $2K + 1$ is the length of imaging lines.

The output of the MPAX beamformer is:

$$Y_{MPAX} = \rho_{MPAX} \times Y_{DAS} \quad (6)$$

The MPAX beamformer can effectively suppress clutter, but its resolution is close to that of DAS.

3. Proposed method

3.1. Signal range-standard deviation factor (SRSF)

The amplitude of the echo signal has the characteristics of large amplitude and obvious volatility on both sides of the strong scattering target. As a result, the standard deviation from both sides of the strong scattering target is larger than that of the speckle background [29]. Based on this characteristic, we propose a signal range-standard deviation factor (SRSF) method to suppress the off-axis signals and thus improve the resolution.

$$SRSF(n) = \left(\max(x(n)) - \min(x(n)) \right) / \sqrt{\sum_{m=1}^M (x_m(n) - \bar{x}(n))^2 / M} \quad (7)$$

where $\bar{x}(n) = (\sum_{m=1}^M x_m(n))/M$ is the mean of $x(n)$.

To prevent the influence of outliers, a mean filter is used in SRSF. The SRSF of off-axis signals around a strong scattering target will be smaller than that of a speckle background. The original SRSF weighted may lead to a decrease in the speckle background quality. Therefore, it is necessary to truncate the SRSF.

$$SRSF_T = \min(SRSF, \eta) \quad (8)$$

where η is the truncate value, which is determined based on the feature that the speckle obtained by DAS beamforming usually follows a Rayleigh distribution [30,31]. In this paper, the range of η is 3.8 to 4.2.

When the echo signal from a strong scattering point is very similar to its surroundings, $SRSF_T$ will erroneously reduce the main lobe energy while suppressing the sidelobes around the strong scattering point. To ensure the mainlobe energy is reserved, the median-filtered SCF is used.

$$\rho_{sSCF}(n) = \text{filter}(\rho_{sSCF}[n - \lambda_d], \dots, \rho_{sSCF}[n], \dots, \rho_{sSCF}[n + \lambda_d]) \quad (9)$$

where $\text{filter}(\cdot)$ is the median filter operation [32], and λ_d is the wavelength. If $\rho_{sSCF}(n)$ is 1, it means that the coherence of the current sampling moment is strong, and $SRSF_T$ is set as η .

The weighting vector of SRSF is defined as follows:

$$\rho_{SRSF}(n) = 10^{(\eta - SRSF_T(n))} \quad (10)$$

Then, the output of SRSF beamformer is obtained as:

$$Y_{SRSF}(n) = \rho_{SRSF}(n) \times Y_{DAS}(n) \quad (11)$$

The SRSF can suppress the sidelobe signals around strong scattering targets, but it cannot suppress the clutter within anechoic cysts. In previous studies, MPAX can effectively improve cyst detection ability. Thus, the SRSF-MPAX beamformer combining MPAX and SRSF is proposed to achieve high resolution and high contrast simultaneously. The framework diagram of the proposed SRSF-MPAX is shown in Fig. 1. The output of SRSF-MPAX beamformer is obtained as follows:

$$Y_{SRSF-MPAX} = \rho_{MPAX} \times \rho_{SRSF} \times Y_{DAS} \quad (12)$$

3.2. The implementation of SRSF-MPAX based on FPGA

This section will provide a detailed introduction to the implementation process of SRSF-MPAX on the FPGA-based ultrasound imaging system to verify its real-time imaging capability.

(1) System Hardware Platform Design

The ultrasound imaging platform designed in this paper mainly consists of the following parts: FPGA circuit, control circuit, front-end ultrasound transmission circuit, AFE echo sampling circuit, USB 3.0 interface, and PC. The overall diagram of the designed ultrasound imaging platform is shown in Fig. 2.

When transmitting ultrasound signals, the PC sends commands to the control processor (TMS320F28335) to configure the parameters in

FPGA. Based on the configured parameters, the FPGA (EP3C10E144I7N) on the transmit board controls the high-voltage switch array (ECN3290TF, 16 tablets) and high-voltage pulse chip (HDL6M05584) to transmit ultrasound waves. Subsequently, low-voltage echo signals, which run through the high-voltage switch array and the TR switch, are sampled by the AFE5805 (4 tablets) at a frequency of 50MHz. Finally, the FPGA (EP4SE360F35I3) performs a series of digital signal processing including digital filtering, dynamic focusing, beamforming, and orthogonal demodulation. The processed data, whether raw echo data or final imaging scan line data, is then uploaded via the USB 3.0 (CYUSB3014) interface. The PC performs logarithmic compression, scan conversion, and image interpolation to display the imaging results. This platform uses line-scan imaging with a 3.5MHz 128-element ultrasound probe that activates 32 sub-elements to form a scan line. All hardware algorithms in this paper are implemented based on Verilog language.

(2) Hardware implementation of the SRSF-MPAX

The scheme of SRSF-MPAX beamformer on FPGA is shown in Fig. 3. A 32-channel parallel adder is used to superimpose the delayed and filtered echo signals for the DAS output, which is the basis of all algorithms. The delayed echo signal should be filtered using a finite impulse response (FIR) filter to remove noise and improve the signal-to-noise ratio. The filtering parameters are configured and stored in FPGA.

The division between range and standard deviation, SCF, and 1-D mean filtering are included in the FPGA implementation of SRSF. The range is mainly obtained using the divide-and-conquer search method to find the maximum and minimum values of the echo data. The standard deviation implementation performs multiplication, summation, and square root operations using the IP cores of LPM-MULT, PAEALL-L-ADD, and ALTSQRT respectively in the FPGA. The division operation is achieved through multiplying and shifting operations. The RAM is used for lookup table to replace exponential calculations in the SRSF algorithm. The calculation process of SCF only involves one square, root, and several addition operations. In FPGA, data exists in binary complement form, and the highest bit of the data is the sign bit (0 for positive numbers and 1 for negative numbers), which can be expressed as

$$\begin{cases} \left(\sum_{m=1}^M \text{sign}(x_m(n)) \right)^2 = |s_n - s_p|^2 \\ s_p + s_n = M \\ s_n = \sum_{m=1}^M \text{high}(x_m(n)) \end{cases} \quad (13)$$

where s_p and s_n is the total number of positive and negative numbers in

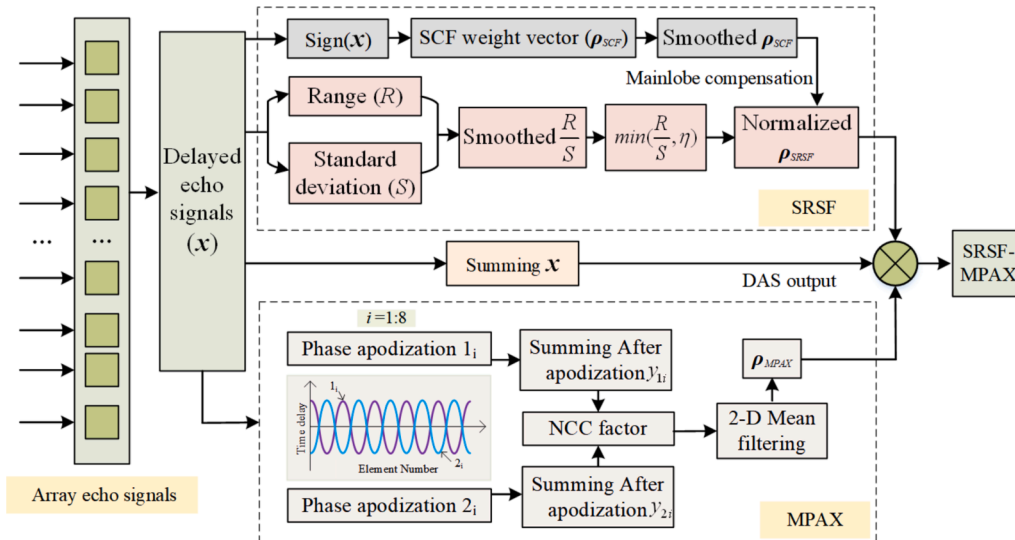


Fig. 1. The framework diagram of the proposed SRSF-MPAX.

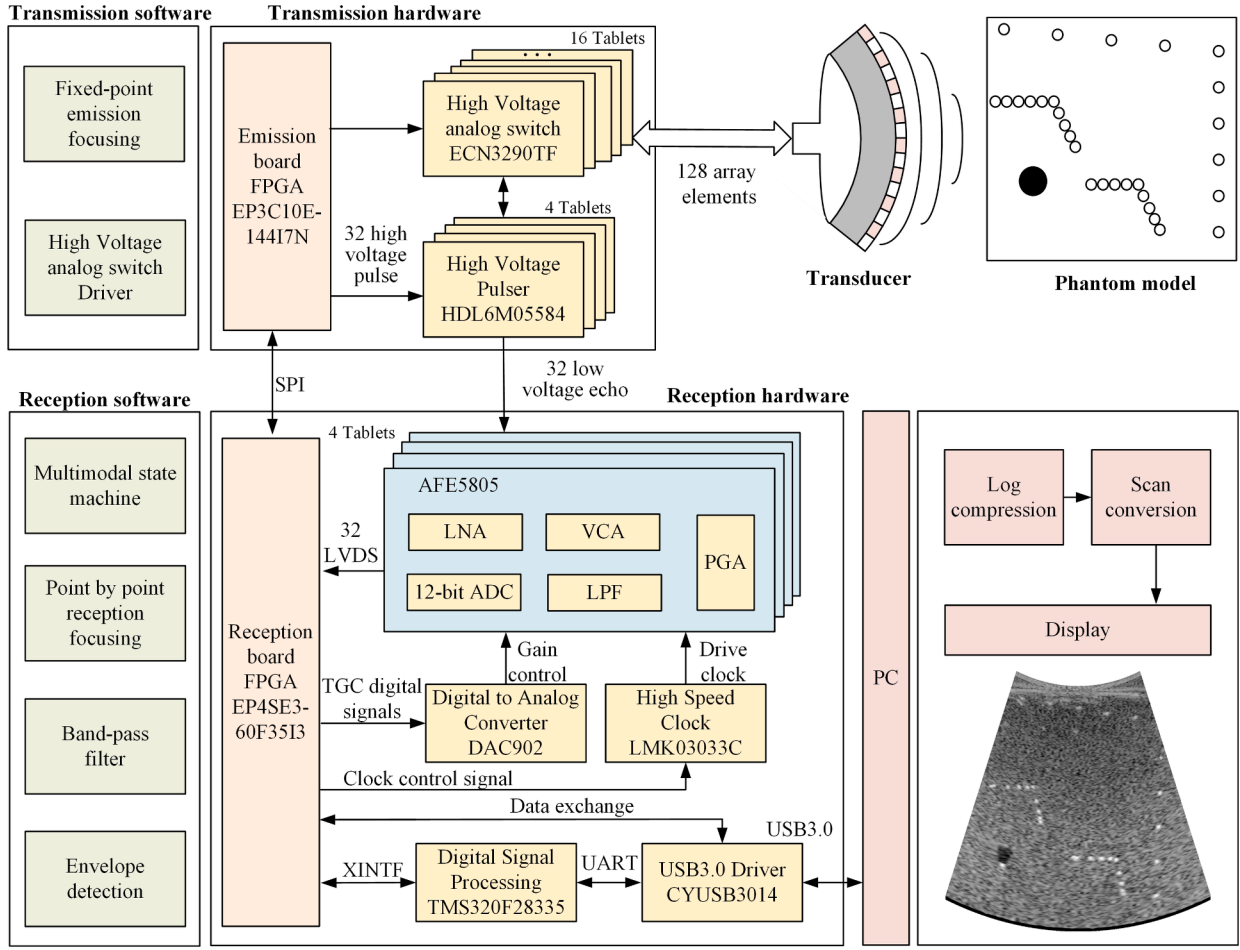


Fig. 2. The overall diagram of the designed open ultrasound imaging system.

the echo signal, respectively. $high(\cdot)$ is the highest bit of the binary complement.

For the implementation of MPAX, it can be divided into three stages: complementary phase apodization, mean NCC calculation, and 2D mean filtering. The phase apodization is to apply a pair of complementary sinusoidal phase apodizations PA_{1i} and PA_{2i} to $x_m(n)$. The signal pairs y_{1i} and y_{2i} used for NCC calculation are:

$$\begin{cases} y_{1i}(n) = \sum_{m=0}^{M-1} x_m(n + PA_{1i}(m)) \\ y_{2i}(n) = \sum_{m=0}^{M-1} x_m(n + PA_{2i}(m)) \\ PA_{1i}(m) = \frac{m_p f_s}{4\pi f_c} \sin(2\pi f_0^i w_d m + \frac{\pi}{2}) \\ PA_{2i}(m) = -PA_{1i}(m) \end{cases} \quad (14)$$

where f_c is the emission frequency, f_s is the sampling frequency, and w_d is the element pitch.

The echo data can be pre-cached in RAM, and then addressed and output based on specific phase apodization values. To avoid floating point operations in the FPGA, the phase apodization value PA_{FPGA} in the FPGA is calculated by Eq.(15).

$$PA_{FPGA} = \text{round}(4 * PA) / 4 \quad (15)$$

where $\text{round}(\cdot)$ is the rounding operation. To save memory resources, the 2-D mean filter only uses the current and previous scanlines data. The phase apodization parameters are listed in Table 1.

4. Results

4.1. Simulated and experimental datasets

Simulation data is generated through Field II software in MATLAB [33]. There are 200,000 scattering points randomly distributed within an imaging depth of 30–50 mm, and an anechoic cyst with a radius of 4 mm is embedded at a depth of 40 mm. The amplitude ratio between the inside and outside of the anechoic cyst is 1000. There are two strong scattering points at a depth of 47 mm with an amplitude 100 times that of the speckle background. The detailed parameter settings are shown in Table 2.

To validate the effectiveness of the proposed SRSF-MPAX, two experimental datasets are used. The public *human heart* dataset (<https://www.ustb.no/ustb-datasets/>) is obtained by scanning the human heart in a parasternal long axis with a Verasonics P4-2v [34]. The *ultrasound phantom* experiment dataset is obtained by acquiring a tissue-mimicking ultrasound phantom (KS107BD), which is manufactured by the Institute of Acoustics Academia Sinica, with our designed open ultrasound imaging system. This paper employs the focused imaging mode for simulation, *human heart* and *ultrasound phantom* experiments. The experimental platform of extracting raw echo data from ultrasound phantom is shown in Fig. 4.

4.2. Evaluation metrics

The evaluation metrics provide a quantitative representation of the differences between beamformers. The full-width at half-maximum (FWHM, -6dB of mainlobe width) is commonly used to measure the

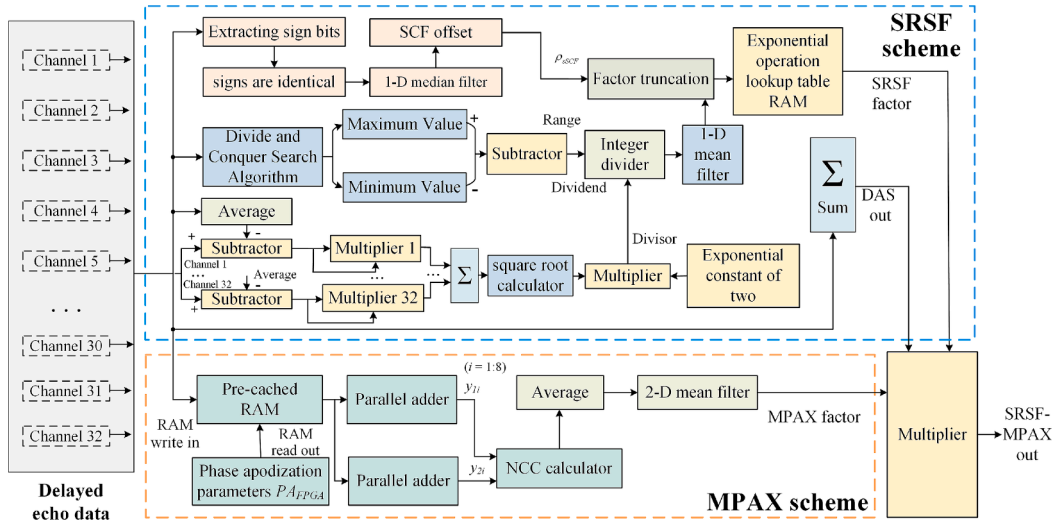


Fig. 3. The scheme diagram of SRSF-MPAX beamformer.

lateral resolution [35]. Contrast ratio (CR) and contrast-to-noise ratio (CNR) are used to assess the effectiveness of cyst imaging, and speckle background quality is evaluated by the speckle signal-to-noise ratio (sSNR) [36]. In addition, the generalized CNR (gCNR) is insensitive as the dynamic range changes, which is beneficial for lesion detection [37,38]. These metrics are calculated by the following equations:

$$CR = \left| 20 \log_{10} \left(\frac{\mu_{ir}}{\mu_{br}} \right) \right| \quad (16)$$

$$CNR = \frac{|\mu_{ir} - \mu_{br}|}{\sqrt{\sigma_{ir}^2 + \sigma_{br}^2}} \quad (17)$$

$$sSNR = \frac{|\mu_{br}|}{\sigma_{br}} \quad (18)$$

$$gCNR = 1 - \int \min\{p_{ir}(x), p_{br}(x)\} dx \quad (19)$$

where μ_{ir} and μ_{br} represent the average values of interest region and background region respectively. σ_{ir} and σ_{br} are the corresponding standard deviation respectively. $p_{ir}(x)$ and $p_{br}(x)$ are the corresponding probability density function respectively. In the following results, the yellow solid areas represent the interest region, and the blue solid areas represent the background regions.

4.3. Anechoic cyst simulation

The results of the anechoic cyst simulation with a dynamic range of 65 dB are shown in Fig. 5. And the lateral variation curves of different beamformers are shown in Fig. 6. As can be seen from Fig. 5(a)-(e), the resolution of SRSF is significantly higher than that of the other beamformers. The resolution of MPAX is almost the same as that of the DAS, but the clutter suppression within the anechoic cysts is significantly improved. The CF and SCF have dark-region artifacts in the speckle background and the speckle uniformity is reduced. The SRSF-MPAX can enhance clutter suppression while improving resolution.

The evaluation metrics with different beamformers are listed in

Table 1
Multiphase apodization parameters.

Parameters	m_p	N	$w_{df0}(\min)$	$w_{df0}(\max)$	Δw_{df0}	ε	A
Phase apodization	4	8	6/M	13/M	1/M	0.001	1λ

Table 3. Compared with DAS, the FWHM of SRSF-MPAX increased by 71.10 %. Compared with SCF, the CNR, and sSNR of SRSF are improved by 128.57 % and 128.21 % respectively. Meanwhile, the gCNR of SRSF-MPAX is improved by 11.17 % compared with SCF.

4.4. Human heart experiment

Fig. 7 displays the results of the human heart experiment in 65 dB dynamic range. It can be seen from Fig. 7(c) that SCF only retains some strong scattering targets in the ventricle while losing other useful tissue information. The SRSF is almost only effective on hyperechoic points to improve resolution. As shown in Fig. 7(f), a clearer intracavitary structure can be obtained by combining MPAX with SRSF while maintaining the resolution. The enlargement of the interventricular septum in the parasternal long-axis views and the corresponding lateral variation curves are shown in Fig. 8, and the specific data is shown in Table 4.

As shown in Table 4, compared with DAS, CF, and MPAX, the FWHM of SRSF-MPAX is reduced by 49.49 %, 24.46 %, and 49.18 % respectively. The CR of SRSF-MPAX is 45.44 dB, which is improved by 43.93 % compared with DAS. The largest improvement in sSNR for SRSF is 163.04 %.

4.5. Ultrasound phantom experiment

The ultrasound phantom experiment results at 65 dB dynamic range are shown in Fig. 9, where the points inside the rectangular area with the tangerine dash are used for the resolution comparison and their lateral variation curves are shown in Fig. 10. According to Fig. 9(b)-(d), it is clear that SCF, CF, and MPAX cause a decrease in tissue brightness in the near-field. The reason is that the depth of emission focus is set to 50 mm based on the positional information of the interest targets, resulting in

Table 2
The detailed parameter settings.

Parameters	Simulation dataset	Human heart dataset	Ultrasound phantom dataset
Sampling frequency (MHz)	50	12	50
Center frequency (MHz)	3.5	3	3.5
Array element numbers (M)	64	64	32
Sound speed (m/s)	1540	1540	1540
Pitch(mm)	0.27	0.3	0.48
Probe	phased array	phased array	convex array

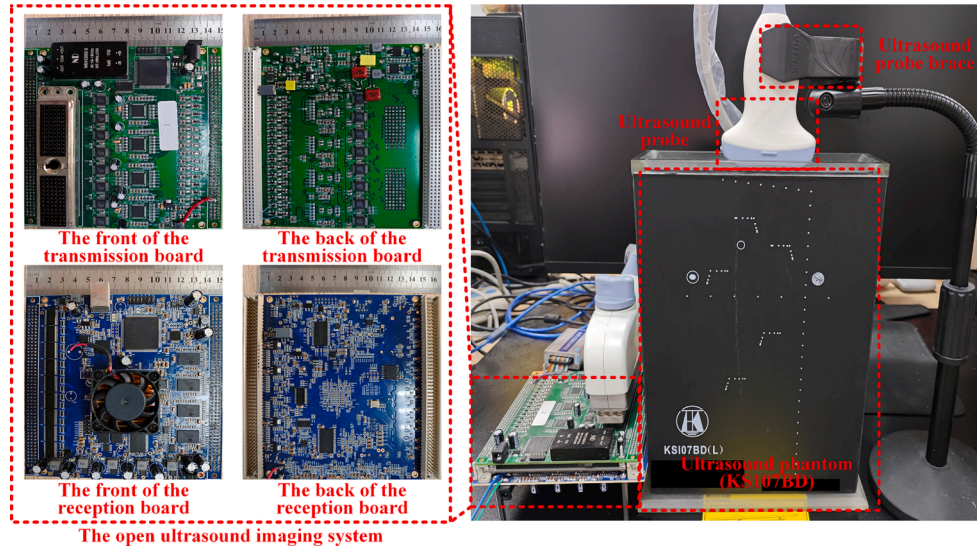


Fig. 4. The experimental platform of extracting raw echo data from ultrasound phantom.

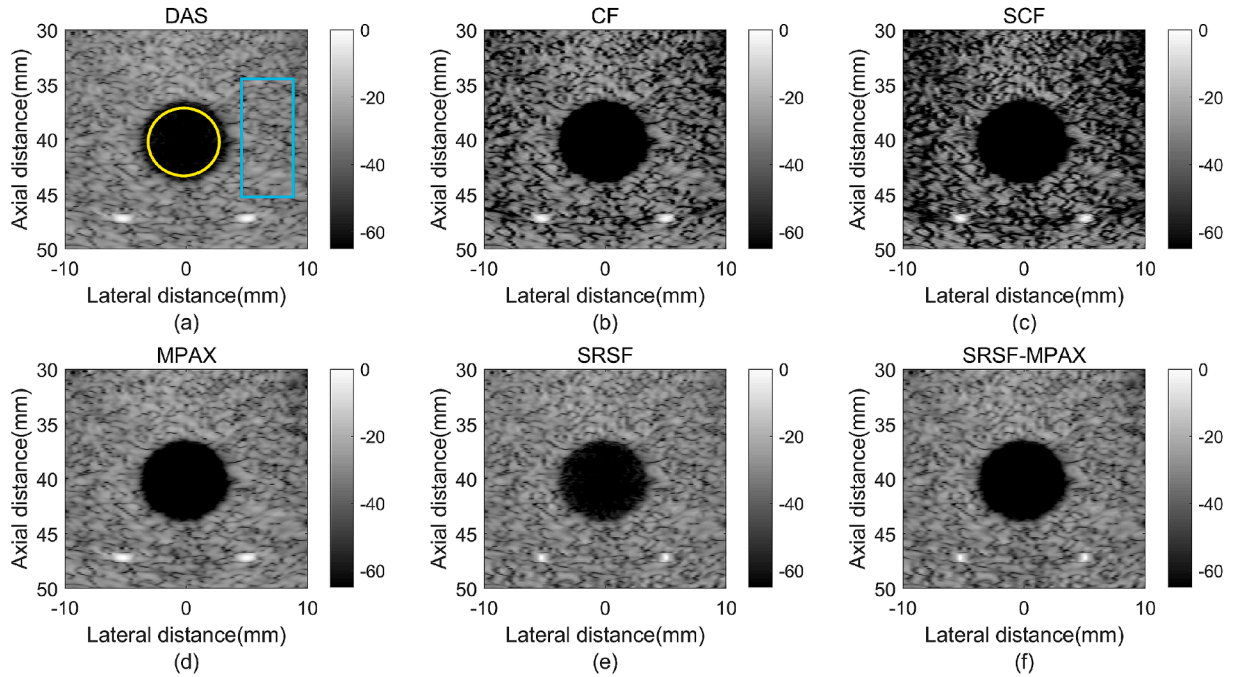


Fig. 5. Simulated anechoic cyst results obtained by (a) DAS, (b) CF, (c) SCF, (d) MPAX, (e) SRSF, (f) SRSF-MPAX.

insufficient near-field energy concentration and reducing the signal-to-noise ratio. However, compared with DAS, the near-field tissue brightness of SRSF is not greatly reduced. It is worth noting that the resolution of SRSF and SRSF-MPAX is improved compared with other beamformers, while SRSF-MPAX can further suppress the clutter.

Specific evaluation metrics for anechoic cysts in Fig. 9 are listed in Table 5. Compared with DAS, the FWHM of SRSF-MPAX is reduced by 63.93 % and the CR of SRSF-MPAX is improved by 53.19 %. The sSNR and CNR of SRSF-MPAX are 97.87 % and 92.55 % higher than those of SCF respectively.

5. Discussion

In this paper, a SRSF beamformer is proposed to improve the imaging resolution. In contrast to the CF beamformer that is prone to generate dark-region artifacts, the SRSF beamformer can selectively suppress

sidelobe artifacts without damaging the speckle background quality based on the standard deviation characteristics of off-axis signals around strong scattering targets. The weighting factors and imaging results for each stage of the SRSF are shown in Fig. 11.

It can be seen from Fig. 11(b) that the ratio of range and standard deviation can identify the sidelobes information around strong scattering points. However, due to its large range of variation in the speckle background, the dark-region artifacts will be introduced by directly weighting the DAS output. Fig. 11(c) illustrates SRSF with truncation threshold, which suppresses sidelobes around strong scattering points while preserving the speckle background well. It is worth noting that a portion of mainlobe lobe of the strong scattering point, located at a depth of 40 mm in Fig. 11(c), has been inaccurately classified as sidelobes and erroneously suppressed. Therefore, we added the median-filtered SCF to preserve the mainlobe.

From the simulation and experiment results, it can be seen that our

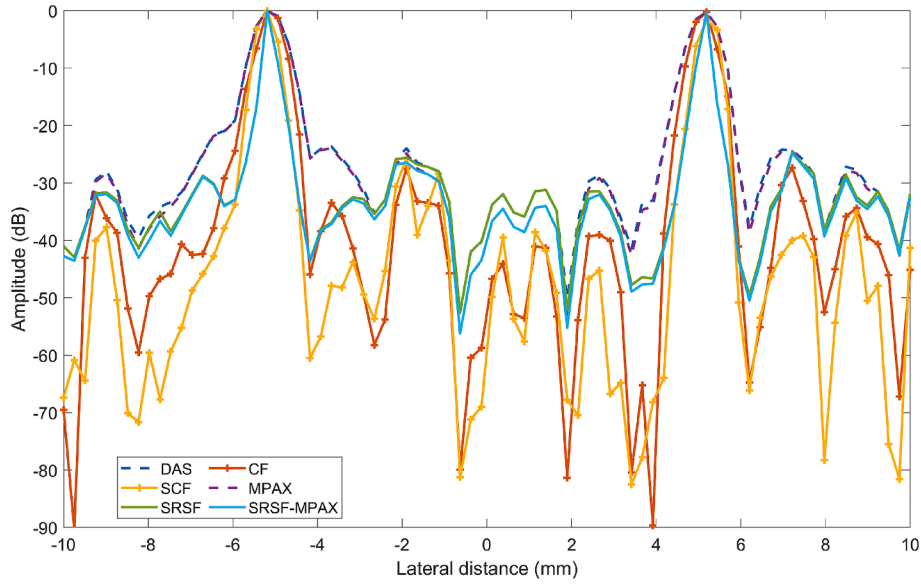


Fig. 6. The lateral variation curves for the simulated anechoic cyst.

Table 3
The metrics of anechoic cyst simulation.

Method	FWHM(mm)	CR(dB)	CNR	sSNR	gCNR
DAS	0.89/0.84	37.46	1.79	1.82	0.9973
CF	0.65/0.61	49.09	1.03	1.04	0.9688
SCF	0.57/0.55	38.68	0.77	0.78	0.8994
MPAX	0.88/0.84	66.71	1.70	1.70	0.9999
SRSF	0.26/0.24	36.98	1.76	1.78	0.9965
SRSF-MPAX	0.26/0.24	56.67	1.67	1.67	0.9999

proposed SRSF-MPAX has the best FWHM. Meanwhile, SRSF-MPAX also has better sSNR and CNR than CF-based algorithms. In the following, the computational complexity of different algorithms is analyzed. The traditional DAS algorithm requires M additions to calculate a single

pixel point. The CF algorithm requires M multiplications and $2M$ additions, which is $3M$ operations in total. The SCF algorithm requires $2M$ additions. The MPAX method requires $4M$ multiplications and $2M$ additions to complete an apodization set and summation, and $(2A + 1)(2K + 1)$ additions for 2D filtering. This apodization and summation needs to be repeated 8 sets, and MPAX requires approximately $8(6M + 3(2A + 1)(2K + 1)) + 1$ operations in total. The SRSF method requires M comparisons, M multiplications and $3M$ additions to calculate the range and standard deviation. Meanwhile, the calculation of the SCF weights also requires $2M$ additions, which is approximately $7M + 1$ operations in total. The SRSF-MPAX method involves calculating SRSF and MPAX separately and then summing them, which is about $55M + 24(2A + 1)(2K + 1) + 2$ operations in total.

In addition to the theoretical complexity analysis, the hardware resource consumption of the FPGA (EP4SE360F35I3) with different

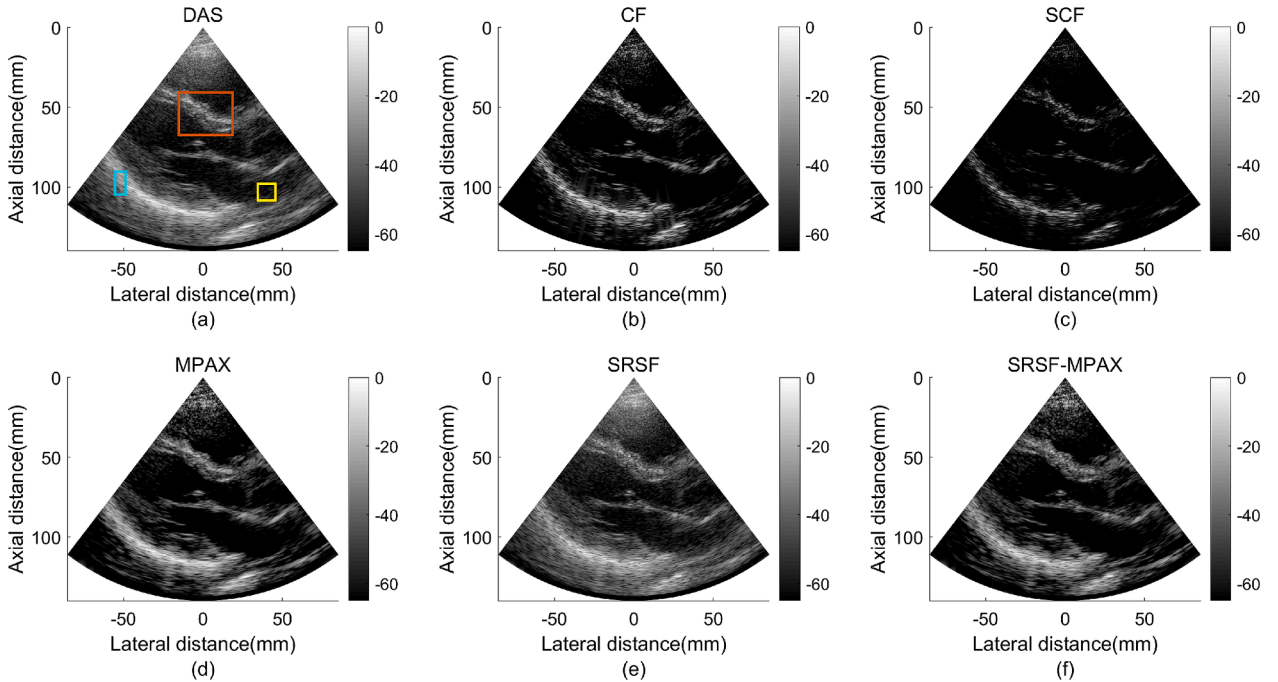


Fig. 7. The results of the human heart experiment by (a) DAS, (b) CF, (c) SCF, (d) MPAX, (e) SRSF, (f) SRSF-MPAX.

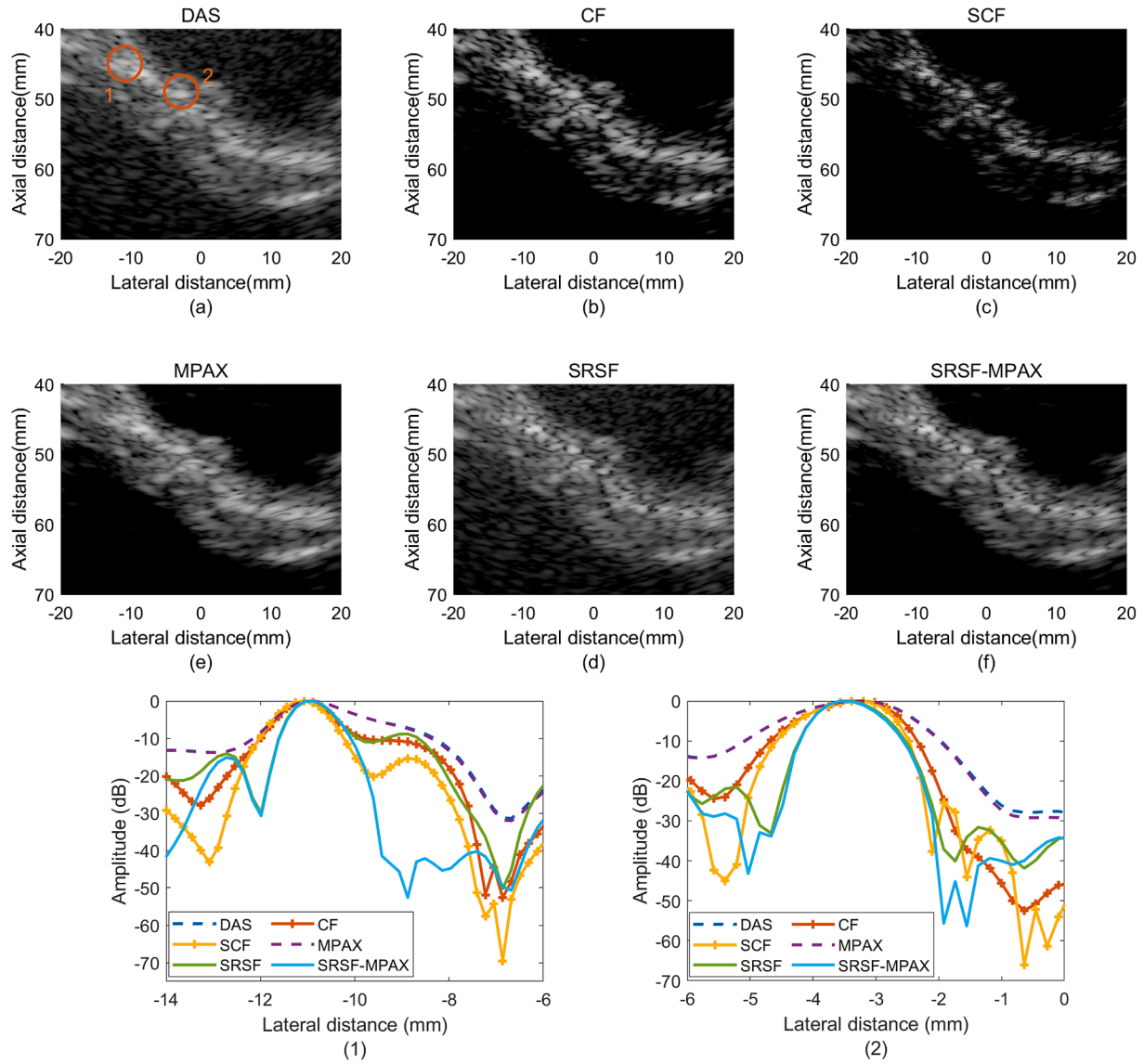


Fig. 8. The enlargement views of the tangerine rectangle by (a) DAS, (b) CF, (c) SCF, (d) MPAX, (e) SRSF, (f) SRSF-MPAX, and the lateral variation curves.

Table 4The evaluation metrics of the *human heart*.

Method	FWHM(mm)	CR(dB)	CNR	sSNR	gCNR
DAS	2.49/2.40	31.57	1.20	1.23	0.9406
CF	1.42/1.85	51.35	0.62	0.62	0.9490
SCF	1.36/1.70	46.90	0.46	0.46	0.9358
MPAX	2.47/2.39	49.56	1.09	1.09	0.9871
SRSF	1.18/1.35	27.67	1.16	1.21	0.9153
SRSF-MPAX	1.17/1.30	45.44	1.07	1.07	0.9871

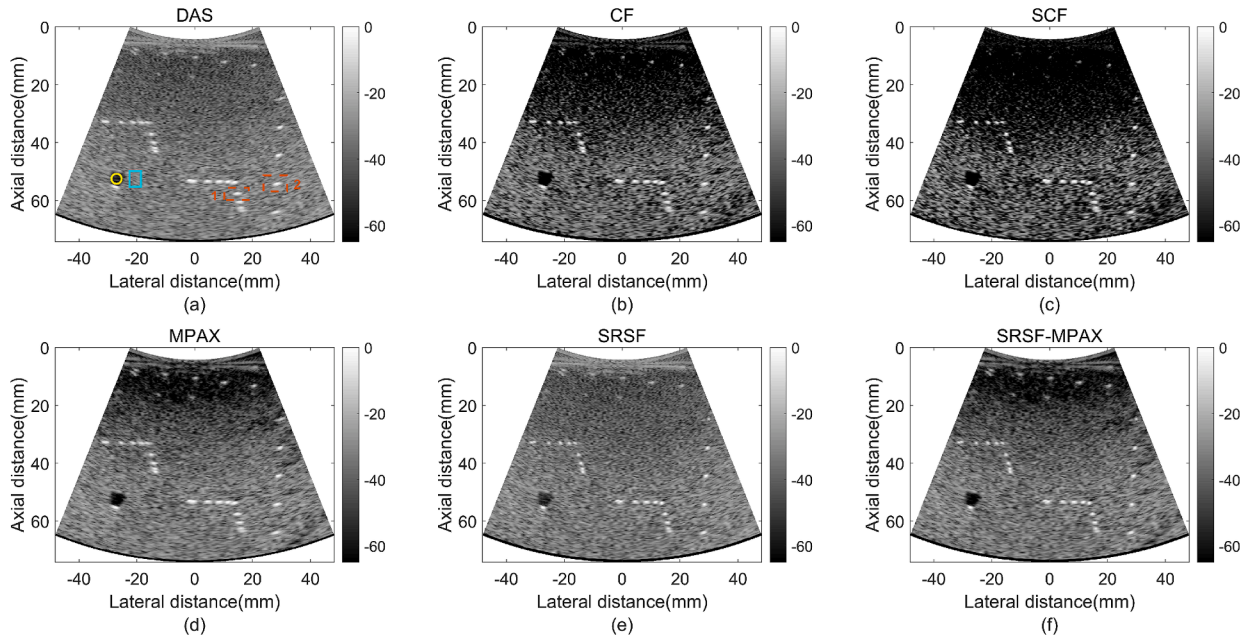
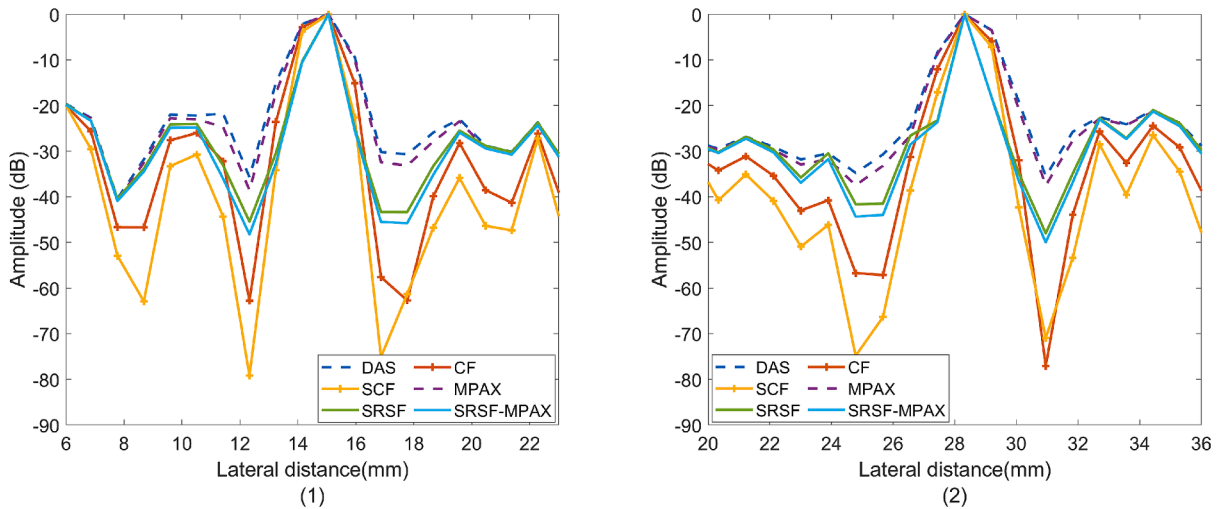
beamformers is shown in Table 6. Among them, the consumption of memory resources is mainly focused on the complete echo data caching, while the DSP resources are predominantly consumed in the 32nd-order FIR filtering of the raw channel echo data. The overall imaging throughput of the SRSF-MPAX beamformer on FPGA is approximately 2289.4MByte/s input and 95.4MByte/s output.

It can be seen in Table 6 that the hardware resource consumption of DAS is the lowest. The hardware resource consumption of CF and SCF is

Table 5The evaluation metrics of the *ultrasound phantom* with different beamformers.

Method	FWHM(mm)	CR(dB)	CNR	sSNR	gCNR
DAS	1.86/1.80	20.40	1.86	2.07	0.9699
CF	1.49/1.43	39.97	1.23	1.25	0.9128
SCF	1.29/1.13	44.37	0.94	0.94	0.9168
MPAX	1.78/1.75	31.56	1.87	1.92	0.9955
SRSF	0.79/0.56	20.07	1.79	1.99	0.9628
SRSF-MPAX	0.77/0.55	31.25	1.81	1.86	0.9929

close to DAS. MPAX requires multiple sets of phase apodizations and NCC calculations for the echo signal, resulting in a significant increase in hardware resource consumption. In contrast, SRSF can effectively improve resolution with low logic resource consumption. SRSF-MPAX combines the advantages of SRSF and MPAX with a further increase in resource consumption. Compared with traditional DAS, CF and SCF, the logic resources consumption of the proposed SRSF-MPAX has increased by 223.53%, while the DSP resource consumption has increased by

**Fig. 9.** The results of the *ultrasound phantom* experiment by (a) DAS, (b) CF, (c) SCF, (d) MPAX, (e) SRSF, (f) SRSF-MPAX.**Fig. 10.** The lateral variation curves for the points within the tangerine rectangle.

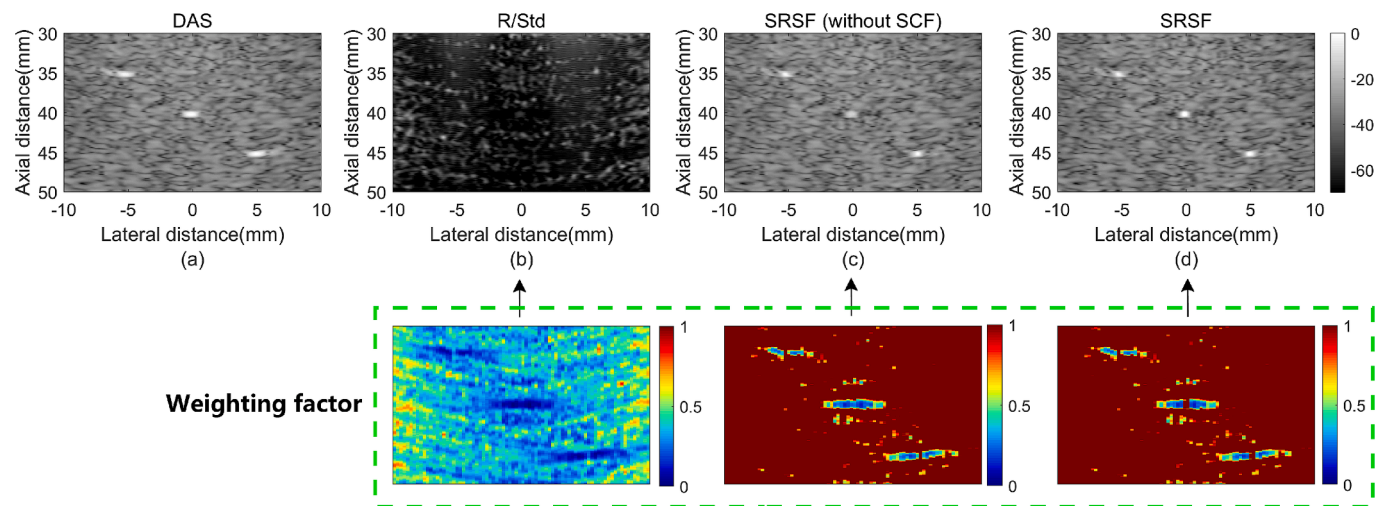


Fig. 11. The weighting factors and imaging results of the SRSF.

Table 6
The hardware resource consumption of the FPGA with different beamformers.

Method	Logic resources	DSP resources	Total block memory bits	Power dissipation
DAS	17 %	82 %	66 %	2.58 W
CF	17 %	88 %	66 %	3.13 W
SCF	17 %	82 %	66 %	3.09 W
MPAX	53 %	90 %	67 %	3.90 W
SRSF	18 %	88 %	66 %	3.21 W
SRSF-MPAX	55 %	96 %	68 %	4.03 W

17.07%, 9.09%, and 17.07%, respectively. The power dissipation of the SRSF-MPAX has correspondingly increased by approximately 56.20%, 28.75%, and 30.42%. Although SRSF-MPAX has increased resource consumption, it has achieved better image quality with limited hardware resources. In the future, the quality of ultrasound images can be further improved by optimizing the hardware platform or improving the beamformer.

6. Conclusion

In this paper, a low-complexity SRSF-MPAX beamformer with high resolution and contrast is proposed, and the corresponding scheme of SRSF-MPAX beamformer on FPGA is introduced in detail. The simulation and *human heart* experiment results show that SRSF-MPAX can achieve better imaging performance. The real-time imaging performance of SRSF-MPAX is consistent with the simulation and experimental dataset results. The performance advantages and acceptable hardware resources consumption of SRSF-MPAX beamformer make it potentially suitable for portable high-quality ultrasound imaging systems.

CRediT authorship contribution statement

Qianwen Li: Writing – original draft, Software, Methodology. **Peng Wang:** Writing – original draft, Software, Data curation. **Kunlin Wang:** Validation, Software, Data curation. **Jinghan Chen:** Writing – original draft, Software, Methodology. **Yue Shen:** Validation, Formal analysis, Data curation. **Ping Wang:** Writing – original draft, Project administration, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The link to the dataset used in the manuscript has been provided.

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